

Course Code: CS2024		
Name of the Programme: B.Tech (Hons.) CSE		
Name of the Course: Object Oriented Programming with Java		
Semester: IV		
Course Credit:	No. of hours per week (L+T+P):	Total no. of Teaching hours:
4	2+1+2	75
Course Objectives		
a. Introduce the fundamentals of Java and core object-oriented programming concepts such as classes, inheritance, and polymorphism.		
b. To promote modular programming through the use of packages, interfaces, and UML design principles.		
c. To equip students with the skills to handle exceptions and implement multithreaded Java programs.		
d. To familiarize students with strings, collections, file handling, and object serialization in Java.		

Syllabus:

Module – 1: Introduction to Object-Oriented Programming	No. of hours
	6
Introduction to Object-Oriented Programming (OOP); Comparison of Procedural, Object-Oriented, and Scripting Languages; History and Architecture of Java; Features of Java; Java Keywords, Data Types, Variables, Operators, Control Statements, and Arrays.	
Module – 2: Classes and Objects	No. of hours
	6
Classes and Objects, Methods, Constructors, Overloading Constructors, this keyword, Garbage Collection, finalize() method. Inheritance, Super keyword, Polymorphism – Method Overloading, Method Overriding, Access Modifiers, Encapsulation and Getter/Setter methods, Data Abstraction, Abstract Classes, final – keyword, method and class.	
Module - 3: Packages and Introduction to UML Class Diagrams	No. of hours
	6
Packages, Creating a Package, Finding Packages and Classpath, Importing Packages, Interfaces, Creating and Implementing Interfaces, Variables in Interfaces, Extending Interfaces, Introduction to UML Class Diagrams, Creating Class Diagrams to Represent Classes, Interfaces, Attributes, Methods, and Relationships.	
Module - 4: Exception Handling Fundamentals	No. of hours
	6
Exception Handling Fundamentals – Exception types, Using try and catch, Multiple catch clauses, Nested try statements, Java’s built-in exceptions, Creating custom exceptions. Multithreaded Programming – Java thread model, Creating a thread, Creating multiple threads, Thread priorities, Synchronization, Inter-thread communication.	

Module - 5: Operators and Operations	No. of hours
	6
String Constructors, String Operations, Wrapper Classes and Autoboxing/Unboxing, The Collection Framework, Collections, List – ArrayList, LinkedList, Set – HashSet, Queue – PriorityQueue, Map – HashMap, TreeMap, Files and Streams – Byte Streams and File I/O, Character Stream – FileReader and FileWriter, Serialization.	

Course Outcomes: After completing the course, the students will be able to:	
a.	Develop Java applications using fundamentals and object-oriented programming concepts such as classes, inheritance, and polymorphism.
b.	Implement modular design using packages, interfaces, and UML class diagrams.
c.	Apply exception handling and multithreading mechanisms to design robust and concurrent Java programs.
d.	Apply string operations, collection framework, file I/O, and serialization in java applications.

Text Books	
1	H. Schildt, <i>Java: The Complete Reference</i> , 12th ed. New Delhi, India: Tata McGraw-Hill, 2022. ISBN: 978-1260463415.
2	P. Deitel and H. Deitel, <i>Java: How to Program (Early Objects)</i> , 11th ed. Boston, MA: Pearson Prentice Hall, 2017. ISBN: 978-0134743356.

Reference Books	
1	E. Balagurusamy, <i>Programming with Java</i> , 7th ed. New Delhi, India: Tata McGraw-Hill, 2019. ISBN: 978-935316234.
2	Y. D. Liang, <i>Introduction to Java Programming</i> , 12th ed. Boston, MA: Pearson Education, 2018. ISBN: 978-0134670942.

Tutorials	[15 Hours]
1	Comparison of Procedural, Object-Oriented, and Scripting Programming Paradigms
2	Java Architecture and Execution Model
3	Access Modifiers and Package-Level Visibility
4	Representation of Object-Oriented Relationships using UML Class Diagrams
5	Java Thread Lifecycle and State Transitions
6	Wrapper Classes and the Autoboxing/Unboxing Mechanism
7	Serialization and Deserialization

Lab Programs	[30 Hours]
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1	Develop basic Java programs using classes, objects, and methods.
2	Demonstrate constructor overloading and use of this keyword.
3	Design and implement different types of inheritance through classes in Java.
4	Demonstrate method overloading and method overriding in a Java application.
5	Implement an abstract class and define its behavior in a concrete subclass.
6	Develop a Java program using interfaces and demonstrate multiple and hybrid inheritance through interface implementation.
7	Create and use user-defined packages across multiple Java files.
8	Handle built-in exceptions using try-catch-finally and multiple catch blocks.
9	Implement user-defined exceptions with appropriate use of throw and throws.
10	Create threads using both Thread class and Runnable interface and execute them concurrently.
11	Demonstrate inter-thread communication using wait(), notify(), notifyAll() and synchronization blocks.
12	Use built-in methods of the String class for string manipulation operations.
13	Implement collections like List, Set, Queue, Map in java.
14	Develop java programs to access and perform various operations in file contents.
15	Implement a real-world use case using various object-oriented concepts in Java.